

QUESTION 1 (15 MARKS) Start a new page

a) (i) Given $f(x)$ is an odd function, show that $\int_{-a}^a f(x)dx=0$ using $x = -t$.

(ii) Hence, evaluate $\int_{-2}^2 x^4(1+\sin^3 x)dx$.

b) (i) Let $I_n = \int_1^e x(\ln x)^n dx, n = 0, 1, 2, 3, \dots$

Using integration by parts, show that $I_n = \frac{e^2}{2} - \frac{n}{2} I_{n-1}, n = 1, 2, 3, \dots$

(ii) The area bounded by the curve $y = \sqrt{x}(\ln x)^2, x \geq 1$, the x -axis and the line $x = e$ is rotated about the x -axis through 2π radians.

Find the exact volume of the solid of revolution so formed.

c) (i) Sketch the curve $f(x) = \frac{x^2 - x - 6}{x - 1}$.

(ii) Hence, sketch the graph of $y^2 = f(x)$.

QUESTION 2 (15 MARKS) Start a new page

a) Find the following:

(i) $\int \frac{e^{-x}}{1+e^x} dx$.

(ii) $\int \frac{x}{\sqrt{1-2x-x^2}} dx$.

b) Evaluate in exact form:

(i) $\int_0^{\frac{\pi}{6}} \frac{d\theta}{9-8\cos^2 \theta}$ using the substitution $t = \tan \theta$.

(ii) $\int_0^{\frac{\pi}{6}} \sec^3 2\theta d\theta$.

c) Using the method of cylindrical shells, find the volume generated by revolving the area bounded by the lines $x = \pm 2$ and the hyperbola

$$\frac{y^2}{9} - \frac{x^2}{4} = 1 \text{ about the } y\text{-axis.}$$

QUESTION 3 (15 MARKS) **Start a new page**

- a) Let α, β, γ be the roots of the cubic equation $x^3 + px^2 + q = 0$, where p and q are real. The equation $x^3 + ax^2 + bx + c = 0$ has roots α^2, β^2 and γ^2 . Find a, b, c as functions of p and q .
- b) (i) Find the complex square roots of $5 - 12i$, expressing your answer in the form $a + bi$, where a and b are real.
- (ii) Hence, solve the equation: $z^2 + 4z - 1 + 12i = 0$
- c) Given that $z = \cos \theta + i \sin \theta$,
- (i) Show that $z^n + z^{-n} = 2 \cos n\theta$ using De Moivre's Theorem.
- (ii) Hence, solve the equation: $2z^4 - z^3 + 3z^2 - z + 2 = 0$.

QUESTION 4 (15 MARKS) **Start a new page**

- a) The sequence U_n , is defined such that
- $$U_{n+2} = 4U_{n+1} - U_n, \quad n \geq 1 \text{ and } U_1 = 2, U_2 = 4.$$
- Prove by mathematical induction that:
- $$U_n = (2 + \sqrt{3})^{n-1} + (2 - \sqrt{3})^{n-1}.$$
- b) $P(\cos \theta, \sin \theta)$ and $Q(\sec \theta, \tan \theta)$ lie on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, respectively as shown.

M and N are the feet of the perpendicular from P and Q respectively to the x -axis. $0 < \theta < \frac{\pi}{2}$, and QP meets the x -axis at K . A is the point $(a,0)$.

- (i) Given $\triangle KPM \parallel \triangle KQN$, show that $\frac{KM}{KN} = \cos \theta$.
- (ii) Hence, show that K has coordinates $(-a,0)$.
- (iii) **Show** that the tangent to the ellipse at P has equation $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$, **and** deduce it passes through N .
- (iv) **Given** that the tangent to the hyperbola at Q has equation $\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$, show that the tangent passes through M .
- (v) Show that the tangents PN , QM and the common tangent at A are concurrent. Find the point of concurrence.

QUESTION 5 (15 MARKS) Start a new page

- a) (i) If $\omega = i - 1$, evaluate the following points $\bar{\omega}$, $i\omega$, $\frac{1}{\omega}$
- (ii) Hence, indicate ω , $\bar{\omega}$, $i\omega$, $\frac{1}{\omega}$ on the Argand diagram.
- b) Sketch the region R in the Argand diagram consisting of the points z for which:

$$|\arg z| < \frac{\pi}{3}, \quad z + \bar{z} < 4, \quad |z| > 2$$
- c) On the Argand diagram, P represents the complex number z , and R the number $\frac{1}{z}$. A square $PQRS$ is drawn in the plane with PR as a diagonal. If P lies on the circle $|z| = 2$,
 - (i) Prove that Q will lie on the ellipse whose equation has the form $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$;
 - (ii) Hence, specify the numerical values for a and b .

QUESTION 6 (15 MARKS) **Start a new page**

- a) The base of a solid is the region bounded by the line $y = 2x$ and the parabola $y = 4x - x^2$ and the x -axis. Cross-sections parallel to the x -axis are right-angled isosceles triangles with the hypotenuse in the base of the solid.

- (i) Show that the volume of the solid is given by:

$$V = \frac{1}{16} \int_0^4 (4 + 2\sqrt{4-y} - y)^2 dy$$

- (ii) Hence, find the exact volume of the solid.

- b) A particle of mass, m kg, moves in a straight line with velocity v metres/second, under a constant force, P Newtons, and a resistance, R Newtons. Initially, the particle has a speed v_0 metres/second. If $R = 5 + 3v$ and $P = 10$.

- (i) Show that velocity, $v = \frac{5}{3}(1 - e^{-\frac{3t}{m}}) + v_0 e^{-\frac{3t}{m}}$.

- (ii) Find the terminal velocity of the particle.

- (iii) When the particle accelerates from v_0 to v_1 , show that the distance travelled, x metres, is given by :

$$x = \frac{m}{9} \left[3(v_0 - v_1) + 5 \ln \left(\frac{5 - 3v_0}{5 - 3v_1} \right) \right].$$

QUESTION 7 (15 MARKS) **Start a new page**

- a) With respect to the x and y axis, the line $x = 1$ is a directrix, and the point $(2,0)$ is a focus of a conic of eccentricity $\sqrt{2}$.

Find the equation of the conic, and sketch the curve indicating its asymptotes, foci, and directrices.

- b) A circular cone of semi vertical angle θ is fixed with its vertex upwards as shown. A particle P , of mass $2m$ kg, is attached to the vertex V by a light inextensible string of length $2a$ metres.

The particle P rotates with uniform angular velocity ω radians/second in a horizontal circle on the outside surface of the cone and in contact with it.

- (i) Find the tension (T) in the string, in Newtons.
- (ii) Find the normal force (N) on P , in Newtons.
- (iii) Show that, for the particle to remain in uniform circular

motion on the surface of the cone, then $\omega < \left[\frac{g}{2a \cos \theta} \right]^{\frac{1}{2}}$

where g is acceleration due to gravity.

- c) A car of mass, m kg, with speed v metres/second travels around a circular track of radius R metres, inclined at angle θ to the horizontal and g is the acceleration due to gravity.

- (i) Show that if there is a tendency for the car to slip that

$$\tan \theta = \frac{v^2}{gR}.$$

- (ii) If the speed of the car is now halved, prove that the sideways frictional force F , on the wheels, exerted by the track is given:

$$F = \frac{3mgv^2}{4\sqrt{v^4 + g^2 R^2}}.$$

QUESTION 8 (15 MARKS) Start a new page

- a) Two particles of mass 4kg and 6kg are attached at either end of a light inextensible string of length 7 metres, which passes through a small vertical frictionless ring R . The heavier particle A hangs vertically at a distance of 4 metres below the ring while the other particle B describes a horizontal circle whose centre is O . Let θ be the acute angle which particle B makes with the vertical.

Find:

- (i) The distance OR and the radius OB , of the horizontal circle.
 - (iii) The angular velocity of B about O in revolutions/minute to 2 decimal places (use $g = 9.8 \text{ m/s}^2$).
- b) In the diagram below, EC and ED are perpendicular to BA and AC at G and H respectively. The lines AC and BD meet at I . Let $\angle ECA = \alpha$.

- (i) Copy the diagram, then show that $FGCH$ is a cyclic quadrilateral.
- (ii) Prove $\triangle BCD$ is isosceles.
- (iii) Prove $\triangle CID \parallel \triangle CDA$.
- (iv) Given that $\triangle CIB \parallel \triangle CBA$ and $AB + AD = 2BC$, prove that $2CI = BD$.

SOLUTIONS TO 4 UNIT TRIAL 2002

Question 1 (15 marks)

a) (i) $\int_{-a}^a f(x) dx = \int_{-a}^0 f(x) dx + \int_0^a f(x) dx \quad \dots \dots \dots \textcircled{1}$

Consider $\int_{-a}^0 f(x) dx$

Let $x = -t$

$dx = -dt$

$= \int_a^0 f(-t) dt$

When $x=0$ $t=0$

When $x=-a$ $t=a$

$= - \int_a^0 f(t) dt \leftarrow \text{since } -f(t) = f(-t) \text{ as } f(x) \text{ is odd}$

$= \int_a^0 f(t) dt$

$= - \int_0^a f(t) dt$

But $- \int_0^a f(t) dt \iff - \int_0^a f(x) dx$

$\therefore$ From $\textcircled{1}$: $\int_{-a}^a f(x) dx = - \int_0^a f(x) dx + \int_0^a f(x) dx$

$= 0$ as required.

(ii) $\int_{-2}^2 x^4 (1 + \sin^3 x) dx$

$= \int_{-2}^2 x^4 + \int_{-2}^2 x^4 \sin^3 x dx$

$= \left[\frac{x^5}{5} \right]_{-2}^2 + 0$

$= \frac{64}{5}$

*

Consider $\int_{-2}^2 x^4 \sin^3 x dx$

Let $f(x) = x^4 \sin^3 x$

$f(-x) = (-x)^4 \sin^3(-x)$

$= x^4 \cdot (-\sin^3 x)$

$= -x^4 \sin^3 x$

$= -f(x)$

$\therefore f(x)$ is odd.

$\therefore \int_{-2}^2 x^4 \sin^3 x dx = 0$

(2)

$$b) (i) I_n = \int_1^e x (\ln x)^n dx, \quad n=0, 1, 2, 3, \dots$$

$$= \left[\frac{x^2}{2} (\ln x) \right]_1^e - \int_1^e \frac{x^2}{2} \cdot n (\ln x)^{n-1} \cdot \frac{1}{x} dx$$

$$= \frac{e^2}{2} - \frac{n}{2} \int_1^e x (\ln x)^{n-1} dx$$

$$= \frac{e^2}{2} - \frac{n}{2} \cdot I_{n-1}$$

$$(ii) V = \pi \int_1^e y^2 dx = \pi \int_1^e x (\ln x)^4 dx = \pi \cdot I_4$$

$$\text{Now } I_4 = \frac{e^2}{2} - 2 I_3$$

$$= \frac{e^2}{2} - 2 \left[\frac{e^2}{2} - \frac{3}{2} I_2 \right]$$

$$= -\frac{e^2}{2} + 3 I_2$$

$$= -\frac{e^2}{2} + 3 \left[\frac{e^2}{2} - I_1 \right]$$

$$= e^2 - 3 I_1$$

$$= e^2 - 3 \left[\frac{e^2}{2} - \frac{1}{2} I_0 \right]$$

$$= -\frac{e^2}{2} + \frac{3}{2} I_0 \leftarrow *$$

$$* I_0 = \int_1^e x dx$$

$$= -\frac{e^2}{2} + \frac{3}{2} \left[\frac{e^2}{2} - \frac{1}{2} \right]$$

$$= \left[\frac{x^2}{2} \right]_1^e$$

$$= \frac{e^2}{4} - \frac{3}{4}$$

$$= \frac{e^2}{2} - \frac{1}{2}$$

$$\therefore V = \pi \left(\frac{e^2}{4} - \frac{3}{4} \right)$$

$$= \frac{\pi}{4} (e^2 - 3) u^3$$

(3)

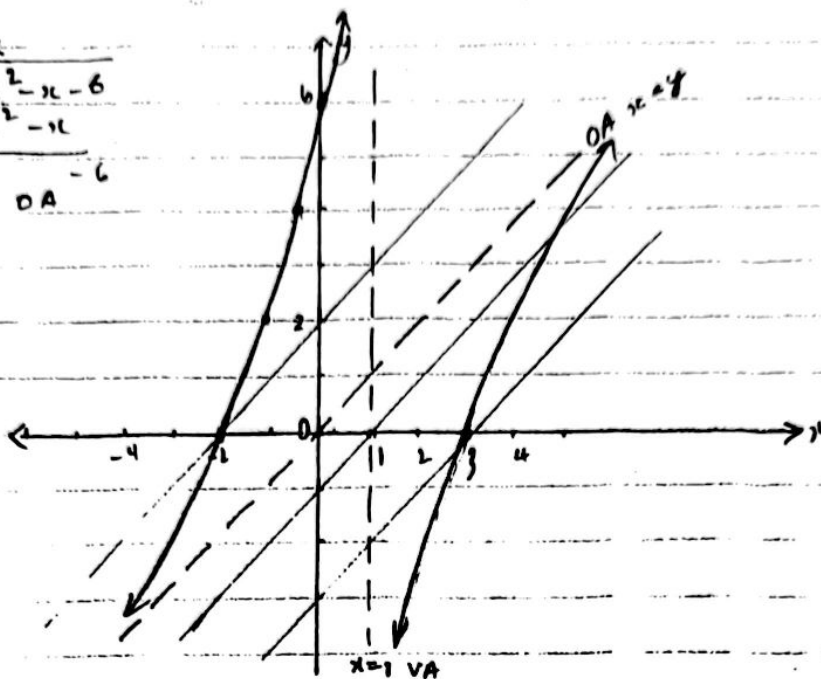
c) (i) $f(x) = \frac{x^2 - x - 6}{x - 1} = \frac{(x - 3)(x + 2)}{x - 1}$

VA: $x = 1$

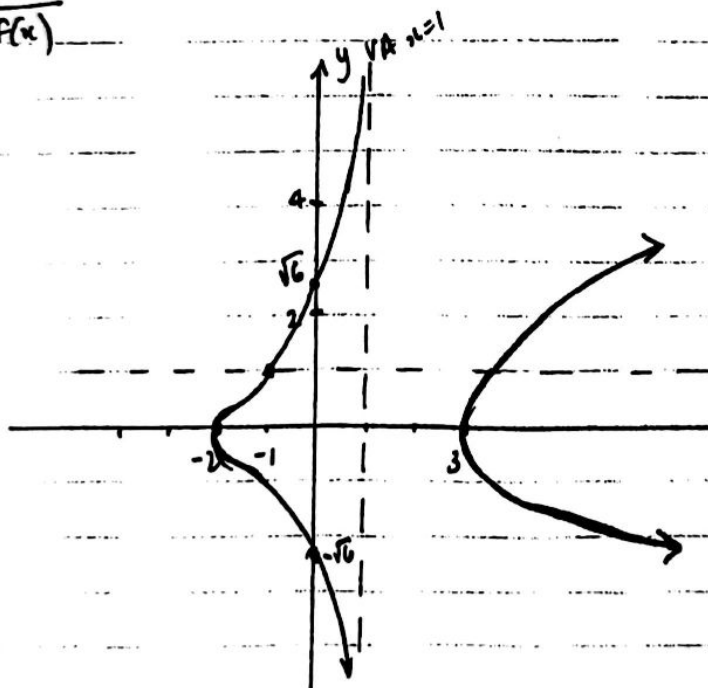
OA: $x - 1 \mid \frac{x^2 - x - 6}{x^2 - x}$

$\therefore y = x$ is OA

y int. $(0, 6)$



(ii) $y^2 = f(x)$
 $y = \pm \sqrt{f(x)}$



(4)

Question 2

a) (i) $\int \frac{e^{-x}}{1+e^x} dx$

$$= \int \frac{1}{e^x(1+e^x)} dx$$

$$= \int \left(\frac{1}{e^x} - \frac{1}{1+e^x} \right) dx$$

$$= \int e^{-x} - \left(\frac{1+e^x - e^x}{1+e^x} \right) dx$$

$$= \int e^{-x} - \left(1 - \frac{e^x}{1+e^x} \right) dx$$

$$= \int e^{-x} - 1 + \frac{e^x}{1+e^x} dx$$

$$= -e^{-x} - x + \ln(1+e^x) + C$$

* Alternative answer

$$\begin{aligned} & \int \frac{1}{1+e^x} dx \\ &= \int \frac{e^{-x}}{e^{-x}+1} dx \\ &= -\ln(e^{-x}+1) \end{aligned}$$

∴ Answer could be:

$$-e^{-x} - \ln(e^{-x}+1) + C$$

(ii) $\int \frac{x}{\sqrt{1-2x-x^2}} dx$

$$= \int \frac{x}{\sqrt{2-(x+1)^2}} dx$$

$$= \int \frac{\sqrt{2} \cos \theta - 1}{\sqrt{2-2\cos^2 \theta}} \cdot -\sqrt{2} \sin \theta d\theta$$

$$= \int \frac{\sqrt{2} \cos \theta - 1}{\sqrt{2(1-\cos^2 \theta)}} \cdot -\sqrt{2} \sin \theta d\theta$$

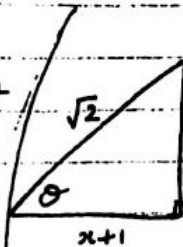
$$= \int \frac{1 - \sqrt{2} \cos \theta}{\sqrt{2} \sin \theta} \cdot \sqrt{2} \sin \theta d\theta$$

$$= \int 1 - \sqrt{2} \cos \theta d\theta$$

$$= \theta - \sqrt{2} \sin \theta + C$$

$$= \cos^{-1} \frac{x+1}{\sqrt{2}} - \sqrt{2} \cdot \frac{\sqrt{1-2x-x^2}}{\sqrt{2}} + C$$

$$= \cos^{-1} \frac{x+1}{\sqrt{2}} - \sqrt{1-2x-x^2} + C$$



*

$$\frac{x+1}{\sqrt{2}} = \cos \theta$$

$$x+1 = \sqrt{2} \cos \theta$$

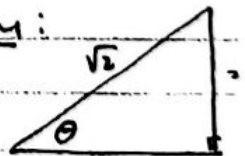
$$dx = -\sqrt{2} \sin \theta d\theta$$

By pythagoras, third side is

$$= \sqrt{1-2x-x^2}$$

$$\therefore \sin \theta = \frac{\sqrt{1-2x-x^2}}{\sqrt{2}}$$

* Alternatively:



$$\frac{x+1}{\sqrt{2}} = \sin \theta$$

$$x+1 = \sqrt{2} \sin \theta$$

$$dx = \sqrt{2} \cos \theta d\theta$$

$$\therefore \cos \theta = \frac{\sqrt{1-2x-x^2}}{\sqrt{2}}$$

* Alternative answer:

(5)

b)

Evaluate exactly:

$$(i) \int_0^{\frac{\pi}{6}} \frac{d\theta}{9 - 8 \cos^2 \theta}$$

using $t = \tan \theta$

$$dt = \sec^2 \theta d\theta$$

$$= (1 + \tan^2 \theta) d\theta$$

$$\therefore d\theta = \frac{dt}{1 + \tan^2 \theta} = \frac{dt}{1 + t^2}$$

$$\theta = 0, t = 0$$

$$\theta = \frac{\pi}{6}, t = \tan \frac{\pi}{6}$$

$$t = \frac{1}{\sqrt{3}}$$

$$= \int_0^{\frac{1}{\sqrt{3}}} \frac{dt}{1 + t^2}$$

$$\frac{9 - 8}{t^2 + 1}$$

Also:



$$\tan \theta = t$$

$$\cos \theta = \frac{1}{\sqrt{t^2 + 1}}$$

$$= \int_0^{\frac{1}{\sqrt{3}}} \frac{dt}{1 + t^2} \cdot \frac{t^2 + 1}{9(t^2 + 1) - 8}$$

$$= \int_0^{\frac{1}{\sqrt{3}}} \frac{dt}{9t^2 + 1}$$

$$= \frac{1}{9} \int_0^{\frac{1}{\sqrt{3}}} \frac{dt}{\frac{1}{9} + t^2}$$

$$= \frac{1}{9} \left[3 \cdot \tan^{-1} 3t \right]_0^{\frac{1}{\sqrt{3}}}$$

$$= \frac{1}{3} \left(\tan^{-1} \sqrt{3} - \tan^{-1} 0 \right)$$

$$= \frac{1}{3} \cdot \frac{\pi}{3}$$

$$= \frac{\pi}{9}$$

$$(ii) \int_0^{\frac{\pi}{6}} \sec^3 2\theta d\theta$$

$$= \int_0^{\frac{\pi}{6}} \sec^2 2\theta \cdot \sec 2\theta d\theta$$

$$= \left[\frac{\tan 2\theta}{2} \cdot \sec 2\theta \right]_0^{\frac{\pi}{6}} - \int_0^{\frac{\pi}{6}} \sec 2\theta \cdot \tan^2 2\theta d\theta$$

$$= \left[\frac{\tan 2\theta \cdot \sec 2\theta}{2} \right]_0^{\frac{\pi}{6}} - \int_0^{\frac{\pi}{6}} \sec 2\theta [\sec^2 2\theta - 1] d\theta$$

$$= - \int_0^{\frac{\pi}{6}} \sec^3 2\theta d\theta + \int_0^{\frac{\pi}{6}} \sec 2\theta d\theta$$

$$\therefore 2 \int_0^{\frac{\pi}{6}} \sec^3 2\theta d\theta = \left[\frac{\tan 2\theta \cdot \sec 2\theta}{2} \right]_0^{\frac{\pi}{6}} + \int_0^{\frac{\pi}{6}} \sec 2\theta d\theta$$

$$u = \sec 2\theta$$

$$du = 2 \sec 2\theta \tan 2\theta$$

$$dv = \sec^2 2\theta$$

$$v = \frac{\tan 2\theta}{2}$$

**

(6)

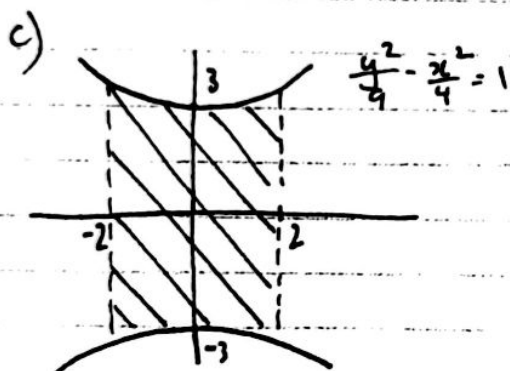
$$= \frac{\sec \frac{\pi}{3} \tan \frac{\pi}{3}}{2} + \int_1^{\frac{\pi}{6}} \sec 2\theta \, d\theta$$

$$= \frac{2}{2} \cdot \sqrt{3} + \frac{1}{2} \ln \left[\sec 2\theta + \tan 2\theta \right]_0^{\frac{\pi}{6}}$$

$$= \sqrt{3} + \frac{1}{2} \ln \left(\sec \frac{\pi}{3} + \tan \frac{\pi}{3} \right) - 0$$

$$= \sqrt{3} + \frac{1}{2} \ln (2 + \sqrt{3})$$

$$\therefore \int_0^{\frac{\pi}{6}} \sec^3 2\theta \, d\theta = \frac{1}{4} \left[2\sqrt{3} + \ln (2 + \sqrt{3}) \right]$$



$$V = 2\pi \int_a^b (\text{shell radius})(\text{shell height}) \, dx$$

$$= 2\pi \int_0^2 x \cdot 2y \, dx$$

$$= 4\pi \int_0^2 xy \, dx$$

$$= \frac{3}{2} \cdot 4\pi \int_0^2 x \sqrt{4+x^2} \, dx \quad *$$

$$= \frac{6\pi}{2} \int_4^8 \sqrt{u} \, du \quad **$$

$$= \frac{2 \cdot 3\pi}{3} \left[u^{\frac{3}{2}} \right]_4^8$$

$$= 2\pi \left(8^{\frac{3}{2}} - 4^{\frac{3}{2}} \right)$$

$$= 2\pi (16\sqrt{2} - 8)$$

$$= 16\pi (2\sqrt{2} - 1)$$

Now $\frac{y^2}{9} - \frac{x^2}{4} = 1$

$$y^2 = 9 \left(1 + \frac{x^2}{4} \right)$$

$$= \frac{9}{4} (4 + x^2)$$

$$y = \frac{3}{2} \sqrt{4 + x^2}$$

Also: Let $u = 4 + x^2$

$$du = 2x \, dx$$

$$\frac{du}{2} = x \, dx$$

If $x = 0$ $u = 4$

$x = 2$ $u = 8$

(7)

Question 3

a) α, β, γ are roots of $x^3 + px^2 + q = 0$
 $\alpha^2, \beta^2, \gamma^2$ " " " $x^3 + ax^2 + bx + c = 0$

b)

Method 1

α, β, γ are the roots of the eqⁿ $x^3 + px^2 + q = 0$ where $\alpha^3 + p\alpha^2 + q = 0$
 $\alpha^2, \beta^2, \gamma^2$ satisfy the required eqⁿ where $y = \alpha^2$
 $\therefore \alpha = \sqrt{y}$

$$\therefore \sqrt{y}^3 + p\sqrt{y}^2 + q = 0$$

$$y^{\frac{3}{2}} + py + q = 0$$

$$y^{\frac{3}{2}} = -py - q$$

$$y^3 = p^2 y^2 + 2pqy + q^2$$

$$y^3 - p^2 y^2 - 2pqy - q^2 = 0$$

This is the same as a polynomial in x :

$$\text{i.e. } x^3 - p^2 x^2 - 2pqx - q^2 = 0$$

$$\therefore \left. \begin{aligned} a &= -p^2 \\ b &= -2pq \\ c &= -q^2 \end{aligned} \right\}$$

Alternate Method

$$\left. \begin{aligned} \alpha + \beta + \gamma &= -p \\ \alpha\beta + \beta\gamma + \gamma\alpha &= 0 \\ \alpha\beta\gamma &= -q \end{aligned} \right\} \text{ using } x^3 + px^2 + q = 0$$

$$\left. \begin{aligned} \alpha^2 + \beta^2 + \gamma^2 &= -a \\ \alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 &= b \\ \alpha^2\beta^2\gamma^2 &= -c \end{aligned} \right\} \text{ using } x^3 + ax^2 + bx + c = 0$$

c)

$$\text{Now: } (\alpha + \beta + \gamma)^2 = \alpha^2 + \beta^2 + \gamma^2 + 2(\alpha\beta + \beta\gamma + \gamma\alpha)$$

$$(-p)^2 = -a + 2 \times 0$$

$$\therefore \underline{a = -p^2} \text{ --- (1)}$$

$$\begin{aligned} \text{Also: } (\alpha\beta + \beta\gamma + \gamma\alpha)^2 &= \alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 + 2\alpha^2\beta\gamma + 2\alpha\beta\gamma^2 + 2\alpha\beta\gamma^2 \\ &= \alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 + 2\alpha\beta(\alpha + \beta + \gamma) \\ 0^2 &= b + 2 \times (-q) \times (-p) \end{aligned}$$

(8)

And: $(z p q)^2 = z^2 p^2 q^2$
 $(-q)^2 = -1$
 $\therefore c = -q^2 \dots \dots \textcircled{3}$

(i) $\sqrt{5-12i} = a+bi$ where a and b are real.

$$5-12i = a^2 + 2abi - b^2$$

$$\therefore \begin{cases} 5 = a^2 - b^2 \\ -6 = ab \end{cases}$$

$$\therefore b = \frac{-6}{a} \Rightarrow a^2 - \frac{36}{a^2} = 5$$

$$a^4 - 5a^2 - 36 = 0$$

$$(a^2 - 9)(a^2 + 4) = 0$$

$$a = \pm 3 \text{ only (as } a \text{ must be real)}$$

$$\therefore \text{If } a = 3 \Rightarrow b = -2$$

$$a = -3 \Rightarrow b = 2$$

$$\therefore \underline{\sqrt{5-12i} = \pm(3-2i)}$$

(ii) $z^2 + 4z - 1 + 12i = 0$

$$z = \frac{-4 \pm \sqrt{16 - 4 \cdot 1 \cdot (12i - 1)}}{2}$$

$$= \frac{-4 \pm \sqrt{16 - 48i + 4}}{2}$$

$$= \frac{-4 \pm 2\sqrt{5-12i}}{2}$$

$$= -2 \pm \sqrt{5-12i}$$

$$\therefore z = -2 + 3 - 2i = \underline{1 - 2i} \quad \text{from (i) above.}$$

$$\text{OR } = -2 - 3 + 2i = \underline{-5 + 2i}$$

(i) $z = \cos \theta + i \sin \theta$

$$z^n = (\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$$

$$z^{-n} = (\cos \theta + i \sin \theta)^{-n} = \cos(-n\theta) + i \sin(-n\theta)$$

$$= \cos n\theta - i \sin n\theta$$

$$\therefore z^n + z^{-n} = \cos n\theta + \cancel{i \sin n\theta} + \cos n\theta - \cancel{i \sin n\theta}$$

$$= \underline{2 \cos n\theta}$$

(9)

$$\begin{aligned}
 (ii) \quad & 2z^4 - z^3 + 3z^2 - z + 2 = 0 \\
 & 2z^4 + 2 - z^3 - z + 3z^2 = 0 \\
 & 2(z^4 + 1) - (z^3 + z) + 3z^2 = 0 \\
 & 2(z^2 + z^{-2}) - (z + z^{-1}) + 3 = 0 \quad \div z^2 \\
 & 2(2\cos 2\theta) - 2\cos \theta + 3 = 0 \\
 & 4\cos 2\theta - 2\cos \theta + 3 = 0 \\
 & 4(2\cos^2 \theta - 1) - 2\cos \theta + 3 = 0 \\
 & 8\cos^2 \theta - 2\cos \theta - 1 = 0 \\
 & (2\cos \theta - 1)(4\cos \theta + 1) = 0 \\
 & \cos \theta = \frac{1}{2} \quad \text{or} \quad \cos \theta = -\frac{1}{4}
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{If } \cos \theta = \frac{1}{2} \quad \sin \theta = \pm \frac{\sqrt{3}}{2} \quad \therefore z = \frac{1}{2} \pm \frac{\sqrt{3}}{2} i \\
 \text{If } \cos \theta = -\frac{1}{4} \quad \sin \theta = \pm \frac{\sqrt{15}}{4} \quad \therefore z = -\frac{1}{4} \pm \frac{\sqrt{15}}{4} i
 \end{aligned}$$

$$\therefore \text{Roots are } z = \frac{1}{2} \pm \frac{\sqrt{3}}{2} i, -\frac{1}{4} \pm \frac{\sqrt{15}}{4} i$$

Question 4

$$2) \quad u_{n+2} = 4u_{n+1} - u_n \quad u_1 = 2; u_2 = 4$$

$$u_n = (2+\sqrt{3})^{n-1} + (2-\sqrt{3})^{n-1}$$

Step 1: Show true for $n=1$: $u_1 = 1+1 = 2$ ✓

$$\text{" " " } n=2: u_2 = (2+\sqrt{3}) + (2-\sqrt{3}) = 4 \quad \checkmark$$

$$\begin{aligned}
 \text{Now } u_3 &= (2+\sqrt{3})^2 + (2-\sqrt{3})^2 \\
 &= 4 + 4\sqrt{3} + 3 + 4 - 4\sqrt{3} + 3 \\
 &= 14
 \end{aligned}$$

* See over

$$\text{Also: } 4u_2 - u_1 = 4 \times 4 - 2 = 16 - 2 = 14$$

$$\therefore u_3 = 4u_2 - u_1$$

$\therefore$ Statement is true for $n=1, n=2, n=3$.

Step 2: Assume statement is true for $n=k, n=k+1$

$$\begin{aligned}
 \text{i.e. } u_k &= (2+\sqrt{3})^{k-1} + (2-\sqrt{3})^{k-1} \\
 u_{k+1} &= (2+\sqrt{3})^k + (2-\sqrt{3})^k
 \end{aligned}$$

Now prove true for $n=k+2$

$$\text{ie Prove: } u_{k+2} = (2+\sqrt{3})^{k+1} + (2-\sqrt{3})^{k+1}$$

(10)

Now $4u_{k+1} - u_k$

$$= 4[(2+\sqrt{3})^k + (2-\sqrt{3})^k] - [(2+\sqrt{3})^{k-1} + (2-\sqrt{3})^{k-1}] \quad (\text{by assumption})$$

$$= (2+\sqrt{3})^{k-1} [4(2+\sqrt{3}) - 1] + (2-\sqrt{3})^{k-1} [4(2-\sqrt{3}) - 1]$$

$$= (2+\sqrt{3})^{k-1} [7+4\sqrt{3}] + (2-\sqrt{3})^{k-1} [7-4\sqrt{3}]$$

$$= (2+\sqrt{3})^{k-1} [4+4\sqrt{3}+3] + (2-\sqrt{3})^{k-1} [4-4\sqrt{3}+3]$$

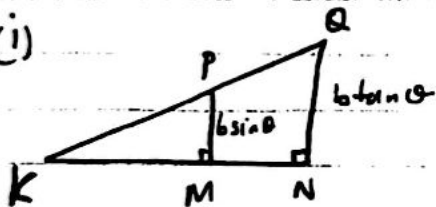
$$= (2+\sqrt{3})^{k-1} (2+\sqrt{3})^2 + (2-\sqrt{3})^{k-1} (2-\sqrt{3})^2 \quad * \text{ from previous page}$$

$$= (2+\sqrt{3})^{k+1} + (2-\sqrt{3})^{k+1}$$

$$= u_{k+2} \quad \text{as required}$$

Step 3: Thus as the result is true for $n=1$, $n=2$ and $n=k+2$ assuming it's true for $n=k$ and $n=k+1$ then it is true for $n=3$, $n=4$ etc and all positive integer n .

b) (i)



By similar triangles:

$$\frac{KM}{KN} = \frac{b \cos \theta}{b \tan \theta}$$

$$\therefore \frac{KM}{KN} = \cos \theta$$

(ii) Let $K = (k, 0)$ Using result above: $KM = a \cos \theta - k$

$$KN = a \sec \theta - k$$

$$\therefore \frac{a \cos \theta - k}{a \sec \theta - k} = \cos \theta$$

$$a \cos \theta - k = (a \sec \theta - k) \cdot \cos \theta$$

$$= a - k \cos \theta$$

$$a \cos \theta - a = k - k \cos \theta$$

$$a (\cos \theta - 1) = k (1 - \cos \theta)$$

$$\therefore \underline{\underline{a = -k}}$$

$\therefore$ K is point $(-a, 0)$ as required

(11)

(iii) $P(a \cos \theta, b \sin \theta)$ is a point on the ellipse $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$.

Now $x = a \cos \theta$

$y = b \sin \theta$

$$\frac{dy}{dx} = \frac{dy}{d\theta} \cdot \frac{d\theta}{dx} = \frac{dy}{d\theta} / \frac{dx}{d\theta}$$

$$\therefore \frac{dy}{dx} = \frac{b \cos \theta}{-a \sin \theta}$$

Eqⁿ of tangent at P is given by:

$$y = \frac{-b \cos \theta}{a \sin \theta} x + B \quad (B \text{ is constant})$$

$$\text{i.e. } b \cos \theta x + a \sin \theta y = C \quad (C \text{ is constant})$$

Since P lies on this tangent, it satisfies eqⁿ above:

$$\text{i.e. } ab \cos^2 \theta + ab \sin^2 \theta = C$$

$$ab(\cos^2 \theta + \sin^2 \theta) = C$$

$$C = ab$$

$\therefore$ tangent at P has eqⁿ:

$$b \cos \theta x + a \sin \theta y = ab$$

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1 \quad (\div ab)$$

as required.

Also, N is point $(a \sec \theta, 0)$.

If the tangent passes through N, its coord. satisfy eqⁿ of tangent:

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$$

$$\text{LHS} = \frac{a/\sec \theta \cos \theta}{a} + 0$$

$$= \frac{1}{\cos \theta} \cdot \cos \theta$$

$$= 1 = \text{RHS}$$

$\therefore$ N lies on the tangent at P.

(iv) M has coord $(a \cos \theta, 0)$ and if lies on the tangent it satisfies the equation $\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$.

$$\text{LHS} = a \cos \theta \cdot \frac{\sec \theta}{a} - 0$$

$$= \cos \theta \cdot \frac{1}{\cos \theta}$$

$$= 1$$

(v) Common tangent at A has equation $x = a$

Required to find the point of intersection of $x = a$ with the tangent at PN

$$\text{i.e. } \left. \begin{aligned} x &= a \\ \frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} &= 1 \end{aligned} \right\}$$

$$\text{i.e. } \cos \theta + \frac{y \sin \theta}{b} = 1$$

$$y = b \frac{(1 - \cos \theta)}{\sin \theta}$$

$\therefore$ PN meets common tangent at $\left(a, \frac{b(1 - \cos \theta)}{\sin \theta} \right)$

Find pt of intersection of $x = a$ with the tangent at QM

$$\text{i.e. } \left. \begin{aligned} x &= a \\ \frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} &= 1 \end{aligned} \right\}$$

$$\text{i.e. } \sec \theta - \frac{y \tan \theta}{b} = 1$$

$$1 - \frac{y \sin \theta}{b} = \cos \theta$$

$$y = b \frac{(1 - \cos \theta)}{\sin \theta}$$

$\therefore$ QM meets common tangent at $\left(a, \frac{b(1 - \cos \theta)}{\sin \theta} \right)$

$\therefore$ PN, QM and tangent at A are concurrent at

$$\left(a, \frac{b(1 - \cos \theta)}{\sin \theta} \right)$$

Question 5

a) (i) $w = i - 1$

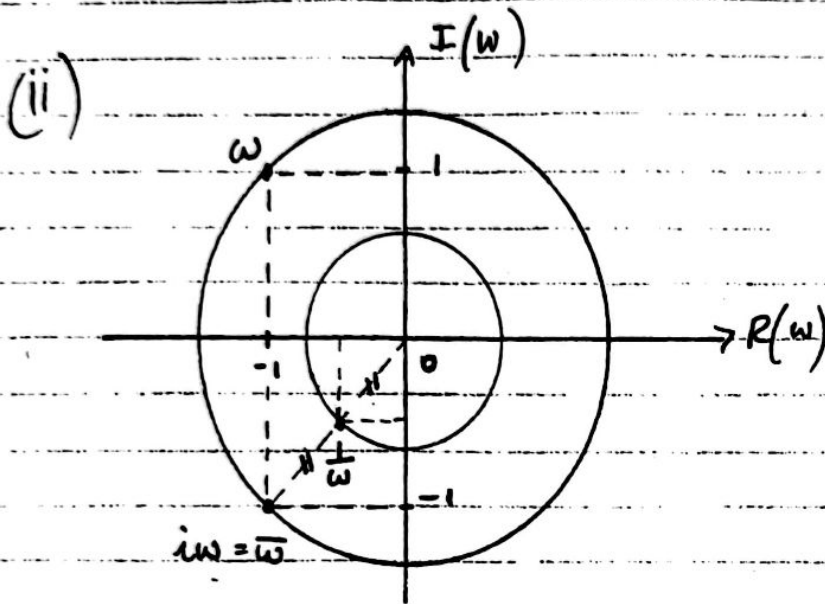
$$\bar{w} = -1 - i$$

$$iw = i(i - 1)$$

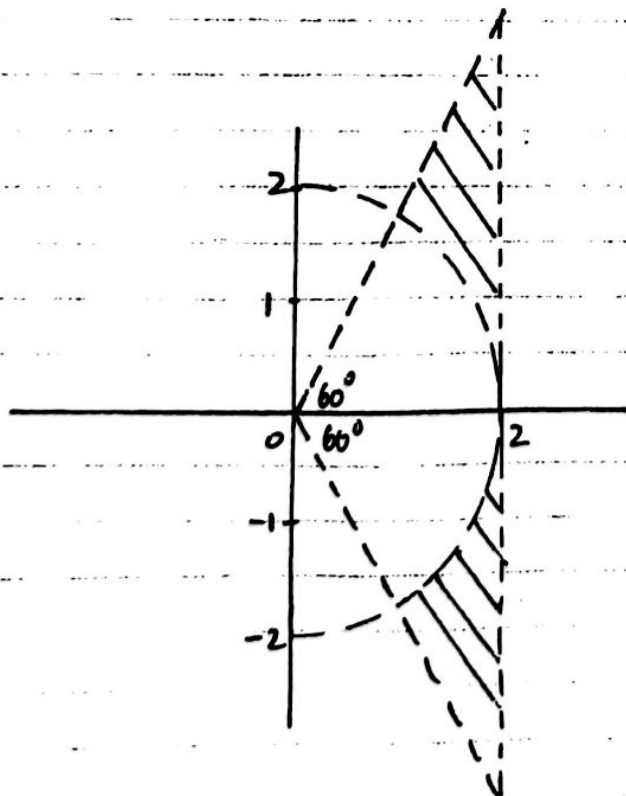
$$= -i + i^2$$

$$= -i - 1 \quad (= \bar{w})$$

$$\frac{1}{w} = \frac{1}{-1-i} = \frac{-1-i}{-1-i} = \frac{-1-i}{2} \quad (= \frac{\bar{w}}{2})$$



b)



$$|z| \geq 2$$

$$z + \bar{z} < 4$$

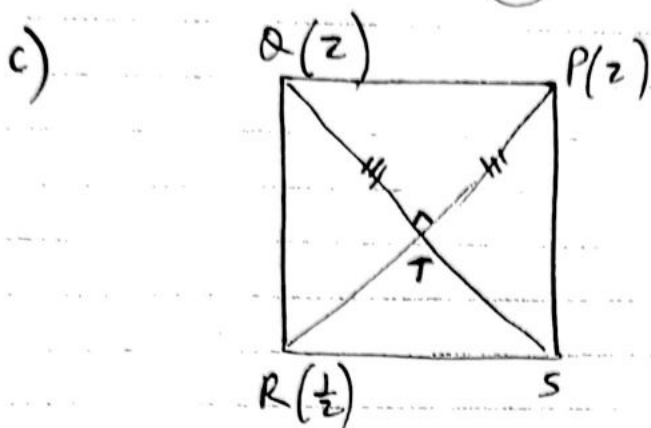
$$x^2 + y^2 \geq 2$$

$$x + yi + x - yi < 4$$

$$2x < 4$$

$$x < 2$$

(14)



We have $P(z)$, $R(\frac{1}{2})$ and $PQRS$ is a square. Let O be point $Q(z)$. Since TQ is obtained from TP by a rotation of $\frac{\pi}{2}$, where T is point $\frac{1}{2}(z + \frac{1}{2})$:

$$z - \frac{1}{2}(z + \frac{1}{2}) = i[z - \frac{1}{2}(z + \frac{1}{2})]$$

$$\therefore 2z = (1+i)z + (1-i)\frac{\bar{z}}{4}$$

By letting $z = 2(\cos \theta + i \sin \theta)$, the eqⁿ above yields:

$$\begin{aligned} 2(x + yi) &= 2(1+i)(\cos \theta + i \sin \theta) + \frac{1}{2}(1-i)(\cos \theta - i \sin \theta) \\ \therefore 2(x + yi) &= \frac{5}{2}(\cos \theta - \sin \theta) + \frac{3i}{2}(\cos \theta + \sin \theta) \end{aligned}$$

hence:

$$\left. \begin{aligned} \frac{4}{5}x &= \cos \theta - \sin \theta \\ \frac{4}{3}y &= \cos \theta + \sin \theta \end{aligned} \right\}$$

$$\left(\frac{4x}{5}\right)^2 + \left(\frac{4y}{3}\right)^2 = \cos^2 \theta - 2\cos \theta \sin \theta + \sin^2 \theta + \cos^2 \theta + 2\cos \theta \sin \theta + \sin^2 \theta$$

$$\frac{16x^2}{25} + \frac{16y^2}{9} = 2$$

$$\frac{x^2}{\frac{25}{8}} + \frac{y^2}{\frac{9}{8}} = 1 \quad \text{which is an ellipse}$$

Alternatively:

$P(z)$ has coordinates $(2\cos \theta, 2\sin \theta)$
 $\therefore OP \equiv z = 2(\cos \theta + i \sin \theta)$

$R(\frac{1}{2})$ has coordinates $(\frac{1}{2}\cos \theta, \frac{1}{2}\sin \theta)$
 $\therefore OR \equiv \frac{1}{2} = \frac{1}{2}\cos \theta - \frac{i}{2}\sin \theta$

$$\begin{aligned} RP &= 2\cos \theta + 2i\sin \theta - \left[\frac{1}{2}\cos \theta - \frac{i}{2}\sin \theta\right] \\ &= \frac{3}{2}\cos \theta + \frac{5}{2}i\sin \theta \end{aligned}$$

$$\begin{aligned} RQ &= \frac{1}{2}[3\cos \theta + 5i\sin \theta]\left[\frac{1+i}{2}\right] \\ &= \frac{1}{4}[3\cos \theta - 5\sin \theta + i(3\cos \theta + 5\sin \theta)] \end{aligned}$$

$$OQ = \frac{5}{4}(\cos \theta - \sin \theta) + \frac{3i}{4}(\cos \theta + \sin \theta)$$

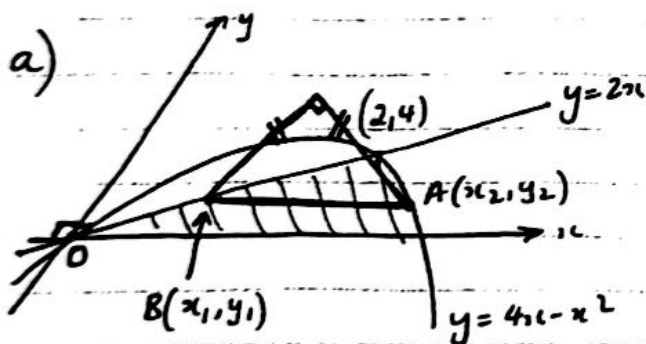
$$\therefore \left. \begin{aligned} x &= \frac{5}{4}(\cos \theta - \sin \theta) \\ y &= \frac{3}{4}(\cos \theta + \sin \theta) \end{aligned} \right\}$$

$$\left. \begin{aligned} \frac{4x}{5} &= \cos \theta - \sin \theta \\ \frac{4y}{3} &= \cos \theta + \sin \theta \end{aligned} \right\}$$

$$\Rightarrow \frac{x^2}{\frac{25}{8}} + \frac{y^2}{\frac{9}{8}} = 1$$

(ii) Eqⁿ of ellipse is $\frac{x^2}{\frac{25}{8}} + \frac{y^2}{\frac{9}{8}} = 1$ where $a = \frac{5}{\sqrt{8}}$
 $b = \frac{3}{\sqrt{8}}$

Question 6



$$\left. \begin{aligned} y &= 2x \\ y &= 4x - x^2 \end{aligned} \right\}$$

Solving gives (0, 0) & (2, 4)
as pts of intersection

Let AB be base of a typical right-angled isosceles triangle

Let $AB = x_2 - x_1$

Now $y = 4x_2 - x_2^2$ at $A(x_2, y_2)$
 $= 4 - (x_2 - 2)^2$

$$\begin{aligned} (x_2 - 2)^2 &= 4 - y \\ x_2 &= 2 \pm \sqrt{4 - y} \end{aligned}$$

$$\therefore x_2 = 2 + \sqrt{4 - y} \quad (\text{since at } x=2, y=4)$$

Also, $y = 2x$, at $B(x_1, y_1)$

$$\therefore x_1 = \frac{y}{2}$$

$$\therefore AB = 2 + \sqrt{4 - y} - \frac{y}{2}$$

(16)

Let 2 equal sides of isosceles triangle be 'a'

By pythagoras' th:

$$2a^2 = \left(2 + \sqrt{4-y} - \frac{y}{2}\right)^2$$



But area of required triangle is $\frac{1}{2} a^2$

$$\therefore A(y) = \frac{1}{4} \left(2 + \sqrt{4-y} - \frac{y}{2}\right)^2$$

$$= \frac{1}{16} (4 + 2\sqrt{4-y} - y)^2$$

$$V = \int_a^b A(y) dy = \frac{1}{16} \int_0^4 (4 + 2\sqrt{4-y} - y)^2 dy$$

as required.

$$\text{ii) } V = \frac{1}{16} \int_0^4 (4-y + 2\sqrt{4-y})^2 dy$$

$$= \frac{1}{16} \int_0^4 \left[(4-y)^2 + 4(4-y)^{3/2} + 4(4-y) \right] dy$$

$$= \frac{1}{16} \left[\frac{(4-y)^3}{-3} + \frac{2 \cdot 4}{5} \frac{(4-y)^{5/2}}{-1} + \frac{4(4-y)^2}{2 \times -1} \right]_0^4$$

$$= -\frac{1}{16} \left[\frac{(4-y)^3}{3} + \frac{8}{5} (4-y)^{5/2} + 2(4-y)^2 \right]_0^4$$

$$= +\frac{1}{16} \left[\frac{64}{3} + \frac{8}{5} \cdot 32 + 32 \right]$$

$$= \underline{\underline{6\frac{8}{15} u^3}}$$

(17)

b)

When $t=0$, $v = v_0$ m/s.

$$R = 5 + 3v$$

$$P = 10$$

$$\begin{aligned} \text{(i)} \quad m\ddot{x} &= P - R \\ &= 10 - (5 + 3v) \\ &= 5 - 3v \\ \ddot{x} &= \frac{5 - 3v}{m} \end{aligned}$$

$$\text{Now } \ddot{x} = \frac{dv}{dt} \Rightarrow \frac{dv}{dt} = \frac{5 - 3v}{m}$$

$$\frac{dt}{dv} = \frac{m}{5 - 3v}$$

$$t = \int \frac{m}{5 - 3v} dv$$

$$t = -\frac{m}{3} \ln(5 - 3v) + C$$

$$\text{When } t=0, v=v_0 \Rightarrow 0 = -\frac{m}{3} \ln(5 - 3v_0) + C$$

$$C = \frac{m}{3} \ln(5 - 3v_0)$$

$$\therefore t = \frac{m}{3} \ln(5 - 3v_0) - \frac{m}{3} \ln(5 - 3v)$$

$$\therefore t = \frac{m}{3} \ln\left(\frac{5 - 3v_0}{5 - 3v}\right)$$

$$\text{Now } \frac{3t}{m} = \ln\left(\frac{5 - 3v_0}{5 - 3v}\right)$$

$$e^{\frac{3t}{m}} = \frac{5 - 3v_0}{5 - 3v}$$

$$(5 - 3v) e^{\frac{3t}{m}} = 5 - 3v_0$$

$$5e^{\frac{3t}{m}} - 3ve^{\frac{3t}{m}} = 5 - 3v_0$$

$$ve^{\frac{3t}{m}} = \frac{5}{3}e^{\frac{3t}{m}} + v_0 - \frac{5}{3}$$

$$v = \frac{5}{3} + v_0 e^{-\frac{3t}{m}} - \frac{5}{3} e^{-\frac{3t}{m}}$$

$$\therefore v = \frac{5}{3} \left(1 - e^{-\frac{3t}{m}}\right) + v_0 e^{-\frac{3t}{m}}$$

(18)

(ii) as $t \rightarrow \infty$, $v \rightarrow \frac{5}{3}$
 $\therefore$ terminal velocity is $\frac{5}{3} \text{ m/s}$.

(iii) $\ddot{x} = \frac{5-3v}{m}$

taking $\ddot{x} = v \cdot \frac{dv}{dx}$ gives:

$$v \cdot \frac{dv}{dx} = \frac{5-3v}{m}$$

$$\frac{dv}{dx} = \frac{5-3v}{mv}$$

$$\frac{dx}{dv} = \frac{mv}{5-3v}$$

$$x = \int_{v_0}^{v_1} \frac{mv}{5-3v} dv$$

$$\therefore x = -\frac{m}{3 \times 3} \int_{5-3v_0}^{5-3v_1} \frac{5-u}{u} du$$

$$= \frac{m}{9} \int_{5-3v_0}^{5-3v_1} \frac{u-5}{u} du$$

$$= \frac{m}{9} \int_{5-3v_0}^{5-3v_1} \left(1 - \frac{5}{u}\right) du$$

$$= \frac{m}{9} \left[u - 5 \ln u \right]_{5-3v_0}^{5-3v_1}$$

$$= \frac{m}{9} \left[5-3v_1 - 5 \ln(5-3v_1) - (5-3v_0) + 5 \ln(5-3v_0) \right]$$

$$= \frac{m}{9} \left[\cancel{5-3v_1} - 5 \ln(5-3v_1) - \cancel{5+3v_0} + 5 \ln(5-3v_0) \right]$$

$$= \frac{m}{9} \left[3(v_0 - v_1) + 5 \ln \left(\frac{5-3v_0}{5-3v_1} \right) \right]$$

Now $u = 5-3v$

$$du = -3 dv$$

$$-\frac{du}{3} = dv$$

$$\text{also: } \frac{5-u}{3} = v$$

When $v = v_0$, $u = 5-3v_0$

$v = v_1$, $u = 5-3v_1$

Question 7

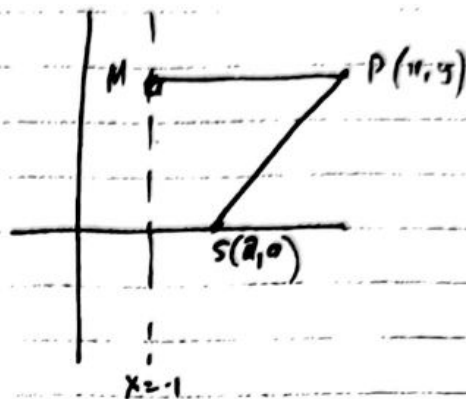
- a) Let $P(x, y)$ be a point on the conic, and let S be pt $(2, 0)$.
 M is the foot of the $\perp$ from P to the directrix.

Then: $PS^2 = e^2 \cdot PM^2$
 $PS^2 = \sqrt{2}^2 \cdot PM^2$ (since eccentricity is $\sqrt{2}$)

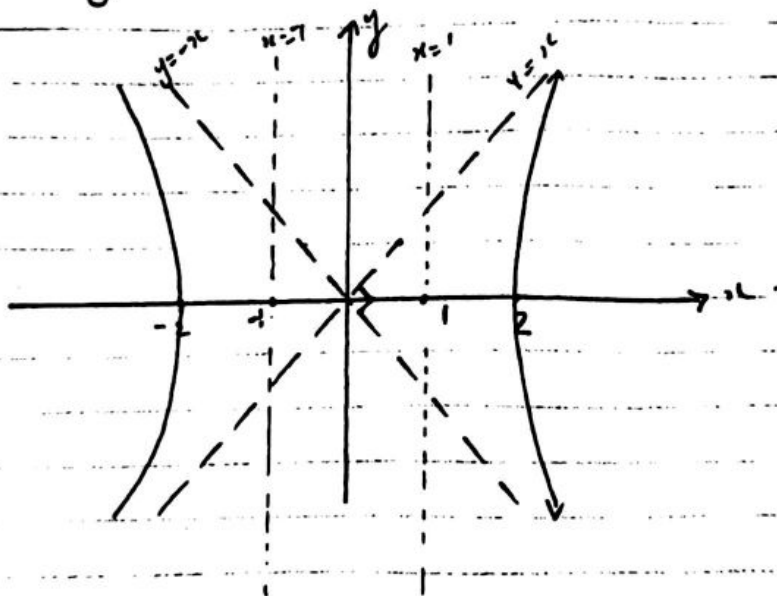
$$(x-2)^2 + y^2 = 2(x-1)^2$$

$$x^2 - 4x + 4 + y^2 = 2x^2 - 4x + 2$$

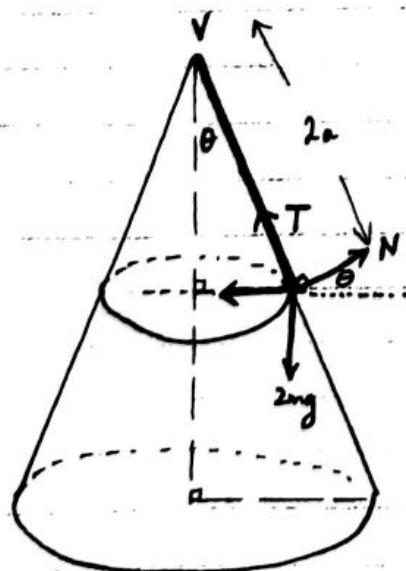
$$\underline{x^2 - y^2 = 2}$$



This is an eqⁿ of a hyperbola
 with asymptotes $y = \pm x$. Since
 the lines are perpendicular to each
 other, the conic is a
rectangular hyperbola.



b)



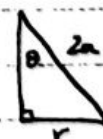
Forces acting on P are:

- ① Centripetal force directed towards the central axis of the cone:

$$= 2mr\omega^2$$

$$= 2m \cdot 2a \cdot \omega^2 \sin\theta$$

$$= 4ma\omega^2 \sin\theta$$



$$\frac{r}{2a} = \sin\theta$$

$$r = 2a \sin\theta$$

- ② Weight = $2mg$

- ③ Normal reaction N , which is $\perp$ to VP



Resolving forces at P (vertically)

$$T \cos \theta + N \sin \theta = 2mg \quad \text{--- (1)}$$

Resolving forces at P (horizontally)

$$T \sin \theta - N \cos \theta = 4ma \omega^2 \sin \theta \quad \text{--- (2)}$$

(i) (1) $\times \cos \theta$ + (2) $\times \sin \theta$:

$$\left. \begin{aligned} T \cos^2 \theta + N \sin \theta \cos \theta &= 2mg \cos \theta \\ T \sin^2 \theta - N \sin \theta \cos \theta &= 4ma \omega^2 \sin^2 \theta \end{aligned} \right\}$$

$$T(\cos^2 \theta + \sin^2 \theta) = 2m(g \cos \theta + 2a \omega^2 \sin^2 \theta)$$

$$\therefore T = 2m(g \cos \theta + 2a \omega^2 \sin^2 \theta) \text{ Newtons}$$

(ii) (1) $\times \sin \theta$ - (2) $\times \cos \theta$:

$$\left. \begin{aligned} T \sin \theta \cos \theta + N \sin^2 \theta &= 2mg \sin \theta \\ T \sin \theta \cos \theta - N \cos^2 \theta &= 4ma \omega^2 \sin \theta \cos \theta \end{aligned} \right\}$$

$$N(\cos^2 \theta + \sin^2 \theta) = 2m(g \sin \theta + 2a \omega^2 \sin \theta \cos \theta)$$

$$\therefore N = 2m(g \sin \theta + 2a \omega^2 \sin \theta \cos \theta) \text{ Newtons}$$

(iii) For the particle to remain in contact with the surface of the cone, then $T > 0$ & $N > 0$ for all values of ω . Since T is always positive, need to only consider N :

$$N > 0 \quad \left[\begin{aligned} \text{ie } g \sin \theta &> 2a \omega^2 \sin \theta \cos \theta \\ g \sin \theta - 2a \omega^2 \sin \theta \cos \theta &> 0 \\ \sin \theta (g - 2a \omega^2 \cos \theta) &> 0 \quad \checkmark \end{aligned} \right]$$

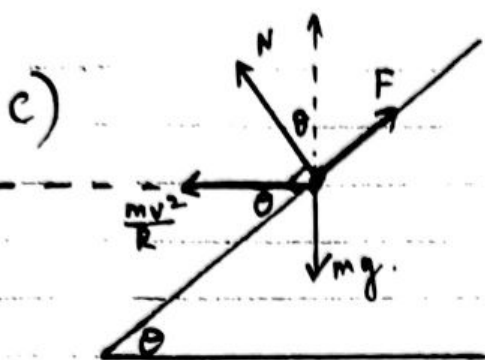
$$\text{ie } g - 2a \omega^2 \cos \theta > 0$$

$$g > 2a \omega^2 \cos \theta$$

$$\frac{g}{2a \cos \theta} > \omega^2$$

$$\omega^2 < \frac{g}{2a \cos \theta}$$

$$\omega < \left(\frac{g}{2a \cos \theta} \right)^{\frac{1}{2}}$$



(i) If there is no lateral force then, $F=0$

$$\text{i.e. } \frac{mv^2}{R} \cos \theta = mg \sin \theta$$

$$\frac{v^2}{Rg} = \frac{\sin \theta}{\cos \theta}$$

$$\frac{v^2}{Rg} = \tan \theta$$

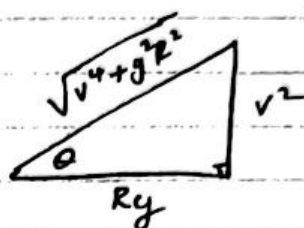
(ii) $F = mg \sin \theta - \frac{mv^2}{R} \cos \theta$

$$= mg \frac{v^2}{\sqrt{v^4 + g^2 R^2}} - \frac{m}{R} \left(\frac{v^2}{2} \right) \frac{Rg}{\sqrt{v^4 + g^2 R^2}}$$

$$= \frac{mv^2 g}{\sqrt{v^4 + g^2 R^2}} - \frac{mv^2 g}{4 \sqrt{v^4 + g^2 R^2}}$$

$$F = \frac{3mv^2 g}{4 \sqrt{v^4 + g^2 R^2}}$$

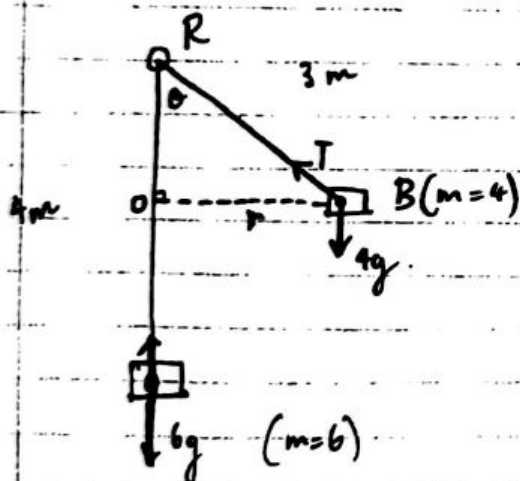
Now since $\tan \theta = \frac{v^2}{Rg}$



$$\sin \theta = \frac{v^2}{\sqrt{v^4 + g^2 R^2}}$$

$$\cos \theta = \frac{Rg}{\sqrt{v^4 + g^2 R^2}}$$

Question 8



As the ring is frictionless, the tensions in AR and BR are equal.

Resolving forces at B gives:

Vertically: $2T \cos \theta = mg$

$$\text{i.e. } T \cos \theta = 4g \quad \text{--- (1)}$$

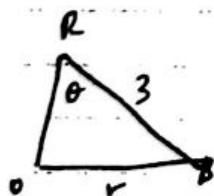
Horizontally: $T \sin \theta = mr\omega^2$

$$\text{i.e. } T \sin \theta = 4 \times 3 \sin \theta \cdot \omega^2$$

$$T = 12\omega^2 \quad \text{--- (2)}$$

Resolving forces at A gives (equilibrium)

$$T = 6g \quad \text{--- (1)}$$

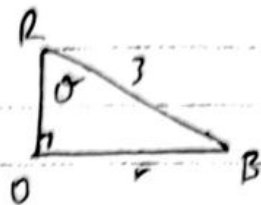


$$\frac{r}{3} = \sin \theta$$

$$r = 3 \sin \theta$$

$$\frac{OR}{3} = \frac{2}{3}$$

$$OR = 2m$$



By Pythagoras, $OB^2 + OR^2 = RB^2$

$$OB^2 + 4 = 9$$

$$OB = \sqrt{5} \text{ m}$$

(ii) From (2): $T = 12\omega^2$ & $T = 6y$

$$6g = 12\omega^2$$

$$\sqrt{\frac{9.8}{2}} = w$$

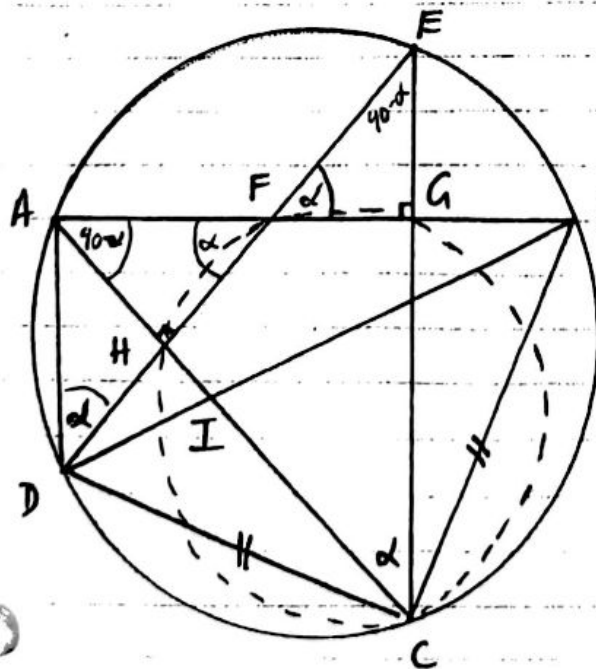
$$\therefore \omega = \sqrt{4.9} \text{ rads/sec}$$

$$= \sqrt{4.9} \cdot \frac{1}{2\pi} \text{ revolutions/sec}$$

$$= \frac{\sqrt{4.9} \times 60}{2\pi} \text{ revolutions/minute}$$

$$\div \underline{21.4 \text{ revolutions/minute (2dp)}}$$

b)



(i) FACH is cyclic

(opposite angles of quad.
FGCH are supplementary)

(ii) Prove $\triangle BCD$ is isosceles.

$$\angle EFB = \alpha \quad (\text{ext. angle of cyclic quad } FGCH \text{ is equal to the interior opposite } \angle GCH)$$

$$\angle AFH = \alpha \quad (\text{vertically opposite angles})$$

$$\angle ADE = \alpha \quad (\text{angles in same segment } AE \text{ of circle (Lye) are equal})$$

$$\angle FAH = 90 - \alpha \quad (\text{angle sum of } \triangle AFH \text{ is } 180^\circ)$$

$$\angle FEG = 90 - \alpha \quad (\text{" " " } \triangle FEG \text{ " "})$$

$$\therefore BC = DC \quad (\text{angles subtended to circumference by chords } BC \text{ \& } CD \text{ are equal})$$

$$\therefore \triangle BCD \text{ is isosceles (2 equal sides } BC \text{ \& } CD)$$

(iii) Prove $\triangle CID \parallel \triangle CDA$

$$\angle DBC = \angle DAC (= \angle DEC) = 90 - \alpha \quad (\text{angles in same segment are equal})$$

$$\angle CDE = \angle DBC = 90 - \alpha \quad (\text{base angles of isosceles } \triangle BCD)$$

$$\therefore \angle CDE = \angle DAC = 90 - \alpha \quad (\angle DAC \text{ is equivalent to } \angle DAE)$$

$$\angle ACD \text{ is common.}$$

$$\therefore \triangle CID \parallel \triangle CDA \quad (\text{equiangular})$$

(iv) Similarly $\triangle CIB \parallel \triangle CBA$ (equiangular)

$$\text{Now } \frac{CI}{DC} = \frac{DI}{AD} \quad (\text{corresponding sides of similar } \triangle CID \text{ \& } \triangle CDA)$$

$$AD = \frac{DC \cdot DI}{CI} \quad \text{--- (1)}$$

$$\text{Also, } \frac{CI}{BC} = \frac{BI}{AB} \quad (\text{corresponding sides of similar } \triangle CIB \text{ \& } \triangle CBA)$$

$$AB = \frac{BC \cdot BI}{CI} \quad \text{--- (2)}$$

$$\textcircled{1} + \textcircled{2}: AB + AD = \frac{BC \cdot BI}{CI} + \frac{DC \cdot DI}{CI}$$

$$2BC = \frac{BC \cdot BI}{CI} + \frac{BC \cdot DI}{CI} \quad (BC = CD)$$

$$2BC = \cancel{BC} \left(\frac{BI + DI}{CI} \right)$$

$$2CI = BD \text{ as required (as } BI + DI = PD)$$